

Program : Diploma in Polymer Technology	
Course Code : 2099	Course Title: CAD Lab - I
Semester : 2	Credits: No Credit
Course Category: Engineering Science	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To provide hands on training on 2D drawing using AutoCAD software.
- To equip the student to be competent in industries for creation and modification of 2D drawings.
- To create an easy platform for students in mastering AutoCAD - 3D.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Mathematics		Mathematics I	I
Orthographic projections - Pictorial & Multi view		Engineering Graphics	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain computer aided drafting and various commands used in AutoCAD	9	Applying
CO2	Draw 2D drawings and sectional views in AutoCAD	17	Applying
CO3	Perform dimensioning and plotting in AutoCAD	12	Applying
CO4	Develop Isometric drawings of simple objects in AutoCAD	5	Applying
	Lab Exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2			3				
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain computer aided drafting and various commands used in AutoCAD		
M1.01	Comprehend Computer Aided Drafting: Applications - Advantages over manual drafting - Hard ware & Soft ware requirements - Different software - Auto CAD - Revit, 3DS MAX, Pro E - IDEAS and Open Source drafting software etc.	1.5	Understanding
M1.02	Familiarize AutoCAD basics - main menu, starting a new drawing, open, save, exit, entering commands using mouse, pull down menu, getting help, data entry, entity selection. Limits of drawing, units, co-ordinates, grid, snap, osnap, ortho mode etc.	1.5	Understanding
M1.03	Locate a point and construct lines - absolute coordinate system-relative coordinate system-polar coordinate system-direct distance entry system.	3	Applying
M1.04	Use Draw commands- lines, circle, arc, ellipse, rectangle, polygon, spline, polyline, helix, donut, block, table, hatch, region etc.	3	Applying
CO2	Draw 2D drawings and sectional views in AutoCAD		
M2.01	Draw simple figures like I - Section, Stanchion mount, Elevation of a building, First & Third angle projection symbols etc.	3	Applying
M2.02	Draw an evenly drilled plate and flange seal	2	Applying
M2.03	Draw orthographic views of an Index plate	2	Applying

M2.04	Draw orthographic views of rod and pipe	2	Applying
M2.05	Draw right half sectional elevation and top view of foot step bearing with bill of materials	3	Applying
M2.06	Use Modify commands - erase, copy, array, rotate, Familiarize AutoCAD basics mirror, offset, scale, move, trim, extend, fillet, chamfer, join, break, extend, stretch, p-line edit, explode etc.	3	Applying
M2.07	Practice Format tools on drawings - Layer, Colour, Line type, Line weight, Transparency, Dimension style, Text style, Multi-leader style etc.	2	Applying
	Lab Exam I	1	
CO3	Perform dimensioning and plotting in AutoCAD		
M3.01	Draw and dimension Template, Slender plate, End bracket etc with captions	3	Applying
M3.02	Draw orthographic views of M10 Hexagonal bolt with dimensions	3	Applying
M3.03	Plotting of Dumb bell tensile sample with detailed dimensions	2	Applying
M3.04	Plotting of Angular tear sample with detailed dimensions	2	Applying
M3.05	Plotting of a typical factory layout with highlighting sizes of various sections	2	Applying
CO4	Develop Isometric drawings of simple objects in AutoCAD		
M4.01	Isometric drawing commands - Isometrip Snap, Isocircle, Direction changing etc.	2	Understanding
M4.02	Isometric view of cube, stepped block, forked end, V- block, angle bracket, shaft support, shaft bearing etc.	3	Applying
	Lab Exam	1	

Text / Reference:

T/R	Book Title/Author
R1	AutoCAD and AutoCAD LT Bible - Ellen Finkelstein (Publisher - John Wiley & Sons Inc.)
R2	AutoCAD Tutorial First Level 2D Fundamentals - Randy H Shih (Publisher - SDC Publications)
R3	Engineering Graphics Essential with AutoCAD instruction - Kirstie Plantenberg (Publisher - SDC Publications)
R4	Engineering Graphics - P I Varghese (Publisher - VIP Publishers)
R5	Geometrical and Machine Drawing - N D Bhatt (Publisher - Charotar Publishing House)

Online Resources:

Sl. No	Website Link
1	https://web.iit.edu
2	www.smarter.com
3	www.notesvarsity.com
4	www.cad-block.com